

CO5 ASSESSMENT TOOL 2(Pseudocode Writing)

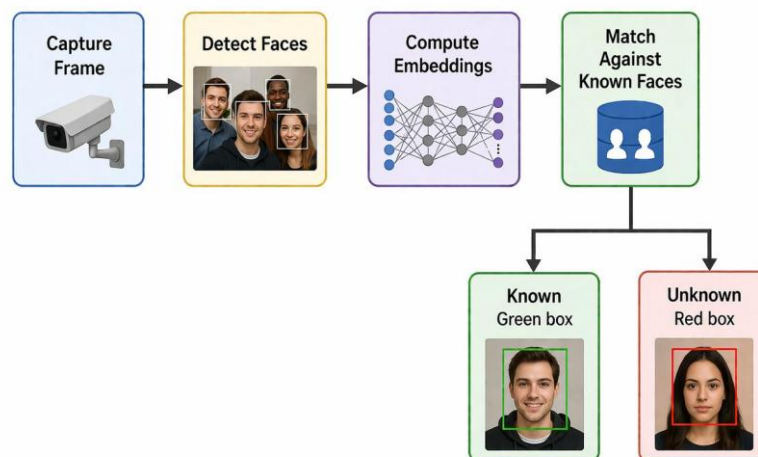
Name: Y.Thanushma

Registration Number: 192425355

Course Code: ITA0502-Computer Vision

1. Real-Time Face Detection and Recognition

- Captures live video frames continuously from the camera.
- Preprocesses the captured frames to improve processing speed.
- Detects multiple faces using an OpenCV-based face detector.
- Extracts facial features from each detected face.



- Compares extracted features with the trained dataset.
- Recognizes registered persons and labels unknown faces as “Unknown.”
- Displays bounding boxes and identity labels on the video.
- Uses an efficient loop to maintain real-time performance and minimize delay

Pseudocode:

BEGIN

// Initialize models and camera

LOAD OpenCV

LOAD face detection model

LOAD face recognition model and trained dataset

OPEN camera

SET recognition threshold

WHILE camera is active DO

// Capture and preprocess frame

 CAPTURE frame

 IF frame is not captured THEN

 BREAK

 END IF

 RESIZE frame

 CONVERT frame to required color format

// Detect multiple faces

 DETECT all faces in frame

 FOR each detected face DO

// Extract facial features

 GET face bounding box

 EXTRACT face region

 RESIZE and NORMALIZE face

 EXTRACT face features

// Recognize the person

 COMPARE features with trained dataset

 IF similarity $\geq$ threshold THEN

 identity $\leftarrow$ matched person's name

 ELSE

 identity $\leftarrow$ "Unknown"

 END IF

// Display result

 DRAW bounding box

 DISPLAY identity label

```

END FOR

// Display processed frame
SHOW frame

// Exit when user presses q
IF key = 'q' THEN
    BREAK
END IF

END WHILE

// Release resources
RELEASE camera

DESTROY windows

END

```

2. Vehicle Detection, Tracking and Counting

How the Constraints Are Satisfied

- Avoid double counting: Each vehicle ID is stored in counted_IDs.
- Multiple vehicles: Each detected vehicle is assigned a separate tracking ID.
- Real-time performance: Frames are resized and only the ROI is processed.
- Vehicle tracking: Vehicle positions are updated across consecutive frames.
- Counting: Vehicles are counted when they cross the virtual line.
- Visualization: Bounding boxes, tracking IDs, and vehicle count are displayed.
- Efficient termination: Pressing q terminates the video loop

Pseudocode:

```

BEGIN

// Initialize models and video

LOAD OpenCV

LOAD vehicle detection model

```

INITIALIZE tracker

OPEN camera/video

DEFINE ROI and virtual counting line

vehicle_count $\leftarrow$ 0

counted_IDs $\leftarrow$ EMPTY SET

WHILE video is active DO

// Capture and preprocess frame

 CAPTURE frame

 IF frame is not captured THEN

 BREAK

 END IF

 RESIZE frame

 SELECT ROI

// Detect vehicles

 DETECT vehicles in ROI

 STORE bounding boxes with confidence above threshold

// Track vehicles across frames

 SEND bounding boxes to tracker

 ASSIGN tracking IDs

 UPDATE vehicle positions

 FOR each tracked vehicle DO

// Check line crossing and avoid double counting

 FIND previous and current center points

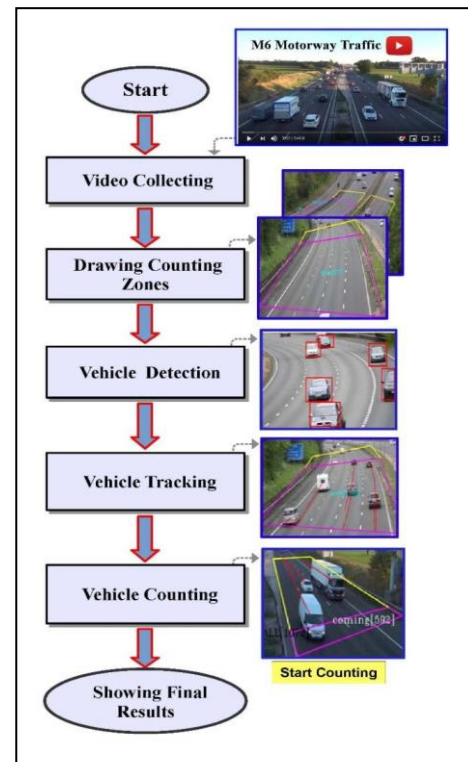
 IF vehicle crosses counting line

 AND ID NOT IN counted_IDs THEN

 vehicle_count $\leftarrow$ vehicle_count + 1

 ADD ID to counted_IDs

 END IF



```
    // Display tracking information

    DRAW bounding box

    DISPLAY tracking ID

END FOR

// Display count and processed video

DRAW counting line

DISPLAY vehicle_count

SHOW frame

// Exit when user presses q

IF key = 'q' THEN

    BREAK

END IF

END WHILE

// Release resources

RELEASE video

DESTROY windows

END
```